

Computer Graphics

Tutorial on display

Course Instructor:
Dr. Suman Kumar Maji

Question 1.

Let the average time to execute an instruction in the display list be $33.33 \mu\text{sec}$. If the frame rate is 30 fps, obtain the max. no. of instructions that may be present in the display list (random scan displays).

Let the average time to execute an instruction in the display list be $33.33 \mu\text{sec}$. If the frame rate is 30 fps, obtain the max. no. of instructions that may be present in the display list (random scan displays).

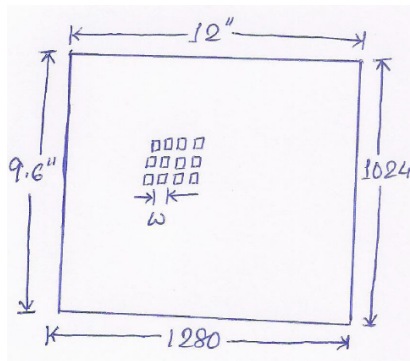
$$\Rightarrow \text{Frame rate} = 30 \text{ Hz.} \quad \text{Time to scan one frame} = \frac{1}{30} \text{ sec} = 33.33 \text{ ms.}$$

$$\text{Maximum no. of instructions} = \frac{33.33 \times 10^{-3}}{33.33 \times 10^{-6}} = 1000$$

Question 2.

Consider the display area of a video monitor to be $12'' \times 9.6''$.
If the resolution of the monitor is 1280×1024 , what is the diameter of each pixel (assume aspect ratio = 1)?

Answer



$$w \times 1280 = 12''$$

$$\Rightarrow w = \frac{12''}{1280} \approx 9.4 \times 10^{-3}''$$

Question 3.

Compute the access time per pixel, for systems with resolutions

(a) 640×480

(b) 1280×1024

Assume a refresh rate of 60fps.

Screen resolution is 640×480 , Refresh rate = 60fps or 60Hz

Time required to scan 1 frame =

$$\frac{1}{60} \text{ sec} \approx 16.7 \text{ ms}$$

So, in 16.7 ms, 640×480 pixels will be accessed.

So, time required to access 1 pixel =

$$\frac{16.7}{(640 \times 480)} \text{ ms} \approx 54 \mu\text{s}$$

Question 4.

Assuming a transfer rate of 0.1 MB/sec, how much time would be necessary to load pix maps with resolutions:

- (a) $512 \times 512 \times 1$
- (b) $1024 \times 1280 \times 1$

Transfer rate = 0.1 MB/sec $\approx$ 102 KB/sec

For 512×512 :

$$\text{Pixmap resolution} = \frac{512 \times 512}{8} \text{ Bytes}$$

$$= \frac{2^9 \times 2^9}{2^3} = 2^5 \times 2^{10} = 32 \text{ KB}$$

$$\text{Time required} = \frac{32}{102} \approx 0.31 \text{ sec}$$

Question 5

A very high resolution graphics display has a Frame Buffer that is 1600 pixels wide and 1280 pixels high. It has non-interlaced 72 Hz refresh rate. Assume 8% of scan line time is required for horizontal retrace and 7% of the frame time (one frame) for vertical retrace.

- (a) Total time required for a single frame, including all retrace?
- (b) Time required for vertical retrace?
- (c) Time required for a single scan line, including 1 horizontal retrace?
- (d) Time required for a single pixel?

Answer a.

(a) Total time required for a single frame, including all retrace?

$$\frac{1}{72} \approx 14ms$$

Answer b.

(b) Time required for vertical retrace?

$$0.07 \times \frac{1}{72} = 0.97ms$$

(c) Time required for a single scan line, including 1 horizontal retrace?

$$\frac{(1 - 0.07) \times 1}{1280 \times 72}$$

(d) Time required for a single pixel?

$$\frac{(\text{answer from c})}{(1 + 0.08) \times 1600}$$